ASCII MASTER FLOW — PANEL 1/12: Design-to-device rail

```
DESIGN / IP / EDA -> architecture, verification and tape-out
FOUNDRY / FAB -> transistor and interconnect layers on wafer
DICING -> wafer becomes individual dies
ATMP / OSAT -> assemble, test, mark and package
BOARD / MODULE / DEVICE -> downstream electronics integration
```

ASCII MASTER FLOW — PANEL 2/12: Semiconductor business-model matrix

```
FAB -> physical wafer-manufacturing plant
FABLESS -> designs and outsources wafer manufacture
FOUNDRY -> manufactures to customers' designs
IDM -> designs and fabricates its own products
RULE -> plant type and firm model answer different questions
```

ASCII MASTER FLOW — PANEL 3/12: Front-end versus back-end

```
FRONT END -> deposition, lithography, etching, doping, metallisation
OUTPUT -> processed wafer containing dies
BACK END -> bonding, encapsulation, testing, marking, packaging
ATMP -> process description; OSAT -> outsourced service model
TRAP -> back-end capacity is not front-end fabrication
```

ASCII MASTER FLOW — PANEL 4/12: Node and lithography ladder

```
MATURE / MID-RANGE -> broad automotive, industrial and power uses
DUV -> mature and mid-range lithography route
LEADING EDGE -> rising complexity and capital intensity
EUV -> concentrated equipment and export-control chokepoint
RULE -> verify every project-specific node claim officially
```

ASCII MASTER FLOW — PANEL 5/12: Advanced integration map

CHIPLETS -> specialised dies combined as one system
2.5D / 3D -> denser package-level integration
THROUGH-SILICON VIAS -> vertical interconnect route
HETEROGENEOUS INTEGRATION -> combine unlike functions or processes
INSIGHT -> performance gains need not begin with smaller lithography

ASCII MASTER FLOW — PANEL 6/12: Materials taxonomy

COMPOUND SEMICONDUCTORS -> GaN / GaAs / InP / SiC
POWER / RF / OPTOELECTRONICS -> application families
SILICON PHOTONICS -> silicon optical-interconnect platform
POLICY GROUPING -> may place unlike technologies together
RULE -> scheme nomenclature does not change material physics

ASCII MASTER FLOW — PANEL 7/12: Fab viability chain

```
RELIABLE POWER + ULTRA-PURE WATER -> stable processing
GASES + CHEMICALS + EQUIPMENT -> controlled process steps
REPEATED RUNS -> yield learning
RELIABILITY EVIDENCE -> customer qualification
OUTCOME -> inauguration alone is not competitive production
```

ASCII MASTER FLOW — PANEL 8/12: Institution and scheme map

```
MeitY -> parent ministry
ISM -> implementing agency
SEMICON INDIA -> scheme package
DLI -> design-focused channel
SPECS / PLI -> separate complementary electronics incentives
```

ASCII MASTER FLOW — PANEL 9/12: Semicon 1.0 to 2.0

```
SEMICON INDIA 1.0 -> Rs 76,000 crore umbrella
FOUR CHANNELS -> fabs, displays, compound/sensor/ATMP, DLI
15 JUL 2026 -> Semicon 2.0 Cabinet approval
SEMICON 2.0 -> Rs 1,27,500 crore approved outlay
SIX PILLARS -> design, tools/materials, fabs, back end, R&D, talent
```

ASCII MASTER FLOW — PANEL 10/12: Facility evidence map

```
DHOLERA -> fab; later named compound-fab plus ATMP approval
MORIGAON -> officially identified OSAT / ATMP location
SANAND -> officially identified back-end cluster and inaugurations
SURAT -> Suchi Semicon OSAT approval
RULE -> location must travel with project and status verb
MUST REMEMBER: The semiconductor chain separates design/IP, wafer fabrication, compound...
```

ASCII MASTER FLOW — PANEL 11/12: Status firewall

APPROVED -> policy permission and support decision
AGREEMENT / GROUNDBREAKING -> financing or construction milestone
INAUGURATED / PRESENTED -> physical or demonstration milestone
QUALIFIED COMMERCIAL SUPPLY -> separate operating evidence
RULE -> never skip a rung or invent production
CLOSE DISTINCTION: Design is not fabrication, fab is not ATMP, packaging is not mere...

ASCII MASTER FLOW — PANEL 12/12: PYQ answer spine

```
DEFINE -> value chain, fab models and ATMP / OSAT
MAP -> ISM, four Semicon 1.0 channels and Semicon 2.0 pillars
ANALYSE -> tools, utilities, talent, yield and qualification
EVIDENCE -> DHRUV64 and facilities with exact status verbs
QUALIFY -> no project total, node, commercial output or answer key
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Date project claims; state value-chain...
```